43 questions | **CLO** 4-A | **Final** 29/12/2026

ANS Full worked answers: Sections 4-3, 4-4 and 4-5

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Each answer shows the set or the rule, the substitution and the value. The `why` lines name the trap the question was built on.

Part A · Multiple choice (16)

Give the letter and a one-line reason, as in a written exam.

A1 4-3

(b)

(4/52)(3/51) = 12/2652 = 1/221. Option (a) is the *with* replacement answer $(4/52)^2$.

A2 4-3

(b)

P(A)·P(B) = 0.2 = P(A ∩ B), so they are independent. They are not mutually exclusive (the intersection is not 0); P(A | B) = 0.2/0.5 = 0.4; P(A ∪ B) = 0.7.

A3 4-3

(b)

P(B | A) = P(A ∩ B)/P(A) = 0.12/0.30 = 0.40. The condition goes in the denominator.

A4 4-3

(b)

If A happens, B certainly does not: P(B | A) = 0 ≠ P(B). Knowing A changes B, so they are dependent.

A5 4-3

(c)

1 − P(no heads) = 1 − $(1/2)^5$ = 31/32.

A6 4-3

(a)

Independent draws: (4/10)(6/10) = 0.24. Option (b), (4/10)(6/9), is the without-replacement value.

A7 4-3

(c)

P(A ∩ B) = 0.3 × 0.6 = 0.18, so P(A ∪ B) = 0.3 + 0.6 − 0.18 = 0.72. Addition rule 2 still applies; independence only gives the intersection.

A8 4-4

(c)

6!/4! = 6 × 5 = 30. Cancel the common 4! instead of expanding both.

A9 4-4

(b)

Repetition allowed: 26 × 26 × 26 × 26 = 26^4 = 456,976.

A10 4-4

(b)

Distinct offices, so order matters: ${}_{10}P_3$ = 10 × 9 × 8 = 720. ${}_{10}C_3$ = 120 would be a committee without titles.

A11 4-4

(a)

Choosing 7 to take is the same as choosing 3 to leave: ${}_nC_r = {}_nC_{n-r}$ = 120.

A12 4-4

(c)

5 letters with L ×2 and E ×2: 5!/(2! 2!) = 120/4 = 30.

A13 4-4

(b)

Fundamental counting rule: 3 × 4 × 2 = 24. "And" across categories means multiply.

A14 4-5

(a)

$_4C_3/_9C_3 = 4/84 = 1/21$. Count both the event and the sample space with combinations.

A15 4-5

(b)

Order doesn't matter, so n(S) = $_{10}C_3 = 120$.

A16 4-5

(b)

With M fixed first, the other 3 letters arrange in 3! = 6 ways, so 6/4! = 6/24 = 1/4. Or directly: each letter is equally likely to be first.

Part B · Short structured questions (16)

Typical 2–6 mark midterm items. Write S and E in set notation.

B1 4-3

(a) Male row: 48/100 = **0.48** (b) Gym column: 48/80 = **0.60** (c) one cell: 48/200 = **0.24**

(d) P(gym) = 80/200 = 0.40, but P(gym | male) = 0.48 ≠ 0.40. Equivalently P(M)·P(G) = 0.5 × 0.4 = 0.20 ≠ 0.24. **Dependent**.

(a) and (b) look alike but have different denominators. The condition chooses the row or column total.

B2 4-3

(a) (5/20)(4/19) = **1/19 ≈ 0.0526**

(b) defective-good *or* good-defective: (5/20)(15/19) + (15/20)(5/19) = **15/38 ≈ 0.3947**

(c) 1 − P(none) = 1 − (15/20)(14/19) = 1 − 21/38 = **17/38 ≈ 0.4474**

(d) 1/19 + 15/38 + 21/38 = 1 ✓

The usual mistake in (b) is to count only one of the two orders.

B3 4-3

Branches: bus 0.6 → late 0.10 / on time 0.90; not bus 0.4 → late 0.04 / on time 0.96.

(a) Multiplication rule 2: 0.6 × 0.10 = **0.06** (b) add both late paths: 0.06 + 0.4 × 0.04 = 0.06 + 0.016 = **0.076** (c) 0.4 × 0.96 = **0.384**

"Of the bus users" makes the 10% a conditional probability, P(late | bus).

B4 4-3

P(A ∩ B) = 0.5 + 0.4 − 0.7 = **0.2**; P(A | B) = 0.2/0.4 = **0.5**; P(B | A) = 0.2/0.5 = **0.4**.

P(A | B) = 0.5 = P(A), so **independent**. Not mutually exclusive, since P(A ∩ B) = 0.2 ≠ 0.

B5 4-3

P(all work) = 0.95^3 = **0.857375**; P(at least one fails) = $1 - 0.95^3$ = **0.142625**.

B6 4-3

The condition B goes in the denominator: P(A | B) = P(A ∩ B)/P(B) = 0.15/0.5 = **0.30**. P(A) is not even given.

B7 4-3

P(A) = 18/36 = 1/2; P(B) = 6/36 = 1/6; A ∩ B = {(2,5), (4,3), (6,1)}, so P(A ∩ B) = 3/36 = 1/12.

P(A)·P(B) = 1/2 × 1/6 = 1/12 = P(A ∩ B): **independent**.

B8 4-4

(a) 10^4 = **10,000** (b) 10 × 9 × 8 × 7 = **5,040** (c) 9 × 9 × 8 × 7 = **4,536**. The first slot has 9 choices (no 0), and the second has 9 because 0 is now allowed but one digit is used.

B9 4-4

(a) ${}_{12}P_3$ = **1,320** (b) ${}_{12}C_3$ = **220** (c) Ahmed is in, choose 2 of the other 11: ${}_{11}C_2$ = **55** (d) ${}_{11}C_3$ = **165**

Check: (c) + (d) = 55 + 165 = 220 = (b) ✓. Every committee either includes Ahmed or doesn't.

B10 4-4

9 letters with A ×2 and T ×2: 9!/(2! 2!) = 362,880/4 = **90,720**.

With S fixed first, arrange the other 8 (A ×2, T ×2): 8!/(2! 2!) = **10,080**.

B11 4-4

(a) 2 men *and* 3 women: ${}_6C_2 \times {}_5C_3 = 15 \times 10 =$ **150**

(b) 4 women + 1 man, or 5 women: ${}_5C_4 \cdot {}_6C_1 + {}_5C_5 = 30 + 1 =$ **31**. "And" multiplies within a case; "or" adds the cases.

B12 4-4

$n(n-1)/2 = 45$, so $n(n-1) = 90 = 10 \times 9$, giving **n = 10**.

B13 4-5

$n(S) = {}_{10}C_4 = 210$. Exactly 2 red: ${}_6C_2 \times {}_4C_2 = 15 \times 6 = 90$, so P = **90/210 = 3/7 ≈ 0.4286**.

At least 1 white = 1 − P(all red) = $1 - {}_6C_4/{}_{10}C_4 = 1 - 15/210 =$ **195/210 = 13/14 ≈ 0.9286**.

B14 4-5

$n(S) = 6!/(3!\,2!) = 60$. Fix A at both ends; arrange B, N, A, N: $4!/2! = 12$.

P = **12/60 = 1/5 ≈ 0.2**

B15 4-5

$n(S) = {}_{10}P_3 = 720$; $n(E) = {}_5P_3 = 60$ (odd digits 1, 3, 5, 7, 9). P = **60/720 = 1/12 ≈ 0.0833**.

B16 4-3 4-5

Combinations: ${}_9C_3/{}_{12}C_3 = 84/220 = 21/55$.

Multiplication rule (without replacement): $(9/12)(8/11)(7/10) = 21/55$. Same answer ✓.

P(none) = **84/220 = 21/55 ≈ 0.3818**; P(at least one) = **136/220 = 34/55 ≈ 0.6182**

Part C · Long questions (5)

Midterm-style, multi-part. Marks in brackets.

C1 4-3 **[12 marks]**

1. 200/400 = **0.5** **2.** 90/150 = **0.6** **3.** 60/200 = **0.3** **4.** 90/400 = **0.225**

5. P(yes | 18–30) = 0.6 ≠ P(yes) = 0.5 (or $0.225 \neq 0.375 \times 0.5 = 0.1875$): **dependent**. Young adults are more likely to exercise.

6. $(100/400)(99/399) =$ **33/532 ≈ 0.062**

C2 4-3 **[10 marks]**

1. A 1/3 → red 3/5, blue 2/5; B 2/3 → red 1/5, blue 4/5.

2. (1/3)(3/5) = **1/5**

3. 1/5 + (2/3)(1/5) = 1/5 + 2/15 = **1/3 ≈ 0.3333**

4. P(A | red) = P(A ∩ red)/P(red) = (1/5)/(1/3) = **3/5 = 0.6**

5. P(A | red) = 0.6 ≠ P(A) = 1/3: **dependent**.

C3 4-4 4-5 **[12 marks]**

1. $_{12}C_5$ = **792**

2. $_7C_3 \times {}_5C_2 = 35 \times 10 = 350$, so P = **350/792 = 175/396 ≈ 0.4419**

3. all engineers $_7C_5 = 21$ or all doctors $_5C_5 = 1$: (21 + 1)/792 = **22/792 = 1/36 ≈ 0.0278**

4. 1 − $_7C_5$/792 = **771/792 = 257/264 ≈ 0.9735**

5. Salim in, choose 4 of the other 11: $_{11}C_4 = 330$, so P = **330/792 = 5/12 ≈ 0.4167**. Check: 5 of 12 seats, 5/12 ✓.

C4 4-4 4-5 **[8 marks]**

1. 2×10^5 = **200,000** **2.** $2 \times 9 \times 8 \times 7 \times 6 \times 5$ = **30,240**

3. $2 \times 10^4 \times 1 / (2 \times 10^5)$ = **0.1** **4.** only 111111 and 222222: 2/200,000 = **0.00001**

C5 4-3 4-4 **[8 marks]**

1. 4^5 = **1024** **2.** $(1/4)^5$ = **1/1024 ≈ 0.000977** **3.** $(3/4)^5$ = **243/1024 ≈ 0.2373** **4.** 1 − 243/1024 = **781/1024 ≈ 0.7627**

Part D · 20-minute mini-quiz (16 marks)

Time yourself, closed book, before opening the answers.

D1 4-3

Independent: the occurrence of one **does not change** the probability of the other (two coin tosses). Dependent: it **does change** it (two cards drawn without replacement).

D2 4-3

Yes: 0.45 = P(motorbike | car). P(car ∩ motorbike) = 0.30 × 0.45 = **0.135**.

D3 4-4

$_8P_3$ = **336**; $_8C_3$ = **56**; 7!/(5! 2!) = **21**

D4 4-4

The middle place is fixed; arrange the other 6: 6! = **720**.

D5 4-5

$_8C_2 \times {}_6C_2 / {}_{14}C_4 = 28 \times 15 / 1001 =$ **420/1001 = 60/143 ≈ 0.4196**

D6 4-3

$P(A \cap B) = 0.6 \times 0.5 =$ **0.30**; $P(B) = 0.30 + 0.4 \times 0.2 =$ **0.38** (add the two tree paths that end in B).

Built to the STAT 2101 Fall 2026 course outline (CLO4-A: calculate probabilities using counting rules and basic probability principles). Sections 4-3 Multiplication Rules and Conditional Probability, 4-4 Counting Rules and 4-5 Probability and Counting Rules only. Every numeric answer was computed by code from the question data.